Class Activity and Notes: Sawyer–Eliassen Balanced Vortex Theory

Tropical Meteorology

Purpose: Explain why concentrated eyewall heating can coexist with subsidence and warming in the eye. Bring the lecture figures to class.

1. Primary versus secondary circulation (8 min)

Define the primary and secondary circulations. Explain which one is approximately in gradient-wind balance.

Complete the balance: $\rho^{-1} p_r =$ _____ + _____. What does Figure 3.2 illustrate?

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2. Read the balanced vortex (8 min)

Using Fig. 3.5, identify (a) where the tangential wind is largest, (b) the vertical distribution of warm anomalies, and (c) the direction in which constant-M surfaces tilt with altitude. Why are temperature and wind shear related?

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3. The Sawyer–Eliassen idea (10 min)

Record the schematic equation $\mathcal{L}[\psi] = \mathcal{F}_Q + \mathcal{F}_F$. Label ψ, N^2, I^2, heating and momentum forcing. Why does a heating source generate flow outside the heated region?

Compare Fig. 3.6a and 3.6b: how does stronger inertial stability change the shape of the response? Compare 3.6b and 3.6c: what does baroclinicity do?

4. Follow the circulation (10 min)

In Fig. 3.7a, trace the main ascent through the heat source and the two compensating return-flow cells. Which direction do M surfaces tilt? Compare this with Fig. 3.7b for momentum forcing.

5. Eyewall heating and eye subsidence (12 min)

Sketch a radial–height section showing eye, eyewall, latent heating, ascent, inner compensating descent, outer subsidence and upper-level outflow. Mark where adiabatic warming occurs. Discuss: Does SE theory guarantee subsidence in every eye at all times? Can an eye remain warm if its mean descent becomes weak? What physical processes are missing from axisymmetric balance theory?

reflection (3 min)

In three sentences, explain why eyewall latent heating can be associated with a clear, warm eye without assuming that the eye itself experiences strong latent heating.

Source: `https://www.meteo.physik.uni-muenchen.de/~roger/TCLecs/Tropical%20Cyclones_Dynamics.html`, sections 3.2–3.8, Figs. 3.2, 3.5–3.7.